

Question Paper

Exam Date & Time: 30-Apr-2025 (02:30 PM - 05:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

FOURTH SEMESTER B.TECH. (ELECTRONICS AND COMMUNICATION ENGINEERING) DEGREE EXAMINATIONS -
APRIL / MAY 2025
SUBJECT: ECE 2222 / ECE_2222 - DIGITAL SIGNAL PROCESSING

Marks: 50

Duration: 180 mins.

Answer all the questions.

- 1A) Perform circular convolution of the following sequences, $x_1(n) = [1, 0, 5, 2]$ and $x_2(n) = [3, 2, 4, 2]$, using DFT-IDFT method. (5)
- 1B) With relevant diagrams and mathematical expressions, describe the concept of overlap-add method. (3)
- 1C) Using Goertzel algorithm, compute $X(2)$ of a sequence, $x(n) = [1, 2, 3, 4]$, for $N=4$. (2)
- 2A) Obtain the 8-point DFT of the sequence $x[n] = [2, 1, 2, 1]$ using radix-2 DIF-FFT algorithm with all the intermediate results in the signal flow graph. (5)
- 2B) The transfer function of a discrete causal system is given by (3)

$$H(z) = \frac{1 - z^{-1}}{1 - 0.2z^{-1} - 0.15z^{-2}}$$

Find difference equation representation and draw the parallel realization.

- 2C) Realize the linear phase FIR filter using the efficient direct form structure having the following impulse response. (2)

$$h(n) = \delta(n) + \frac{1}{4} \delta(n-1) - \frac{1}{8} \delta(n-2) + \frac{1}{4} \delta(n-3) + \delta(n-4)$$

- 3A) Determine the coefficients of a low pass linear phase FIR filter of length $M=15$, that satisfies the condition, $H_r\left(\frac{2\pi k}{15}\right) = \begin{cases} 1, & k = 0, 1, 2, 3 \\ 0.3, & k = 4 \\ 0, & k = 5, 6, 7 \end{cases}$ (5)
- 3B) Determine the system function of a linear phase FIR filter whose length of the impulse response is 5 and one of the locations of the zeros is at $z = 0.5e^{j2\pi/3}$. Also determine the impulse response of this system. (3)
- 3C) Using the filter coefficients obtained in Q3A, realize the filter in efficient direct form. (2)
- 4A) (5)

The specification of desired low-pass Butterworth filter is given as,

$$0.9 \leq |H(e^{j\omega})| \leq 1, \quad 0 \leq \omega \leq \pi/2$$

$$|H(e^{j\omega})| \leq 0.2, \quad 3\pi/4 \leq \omega \leq \pi$$

Determine the analog passband edge frequency (Ω_p), analog stopband edge frequency (Ω_s), order and poles of Butterworth filter required to meet the above filter specification. Use bilinear transformation. Assume sampling interval $T = 1$ sec.

- 4B) For the filter specification given in Q4A, determine analog transfer function $H(s)$ of the corresponding Butterworth filter. (3)
- 4C) For the filter specification given in Q4A, determine the system function $H(z)$ using bilinear transformation. Assume sampling interval $T = 1$ sec. (2)
- 5A) Realize a frequency sampling structure for an FIR causal system whose impulse response is $h(n) = \{1, 2, 3, 4\}$. (5)
- 5B) For the analog transfer function $H(s) = \frac{2}{(s+1)(s+2)}$ find $H(z)$ using impulse invariance method. Assume $T = 0.5$ sec. (3)
- 5C) Explain the periodogram method of power spectrum estimation. Highlight the limitations of this method. (2)

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Event Name		Subject	
MIT MPL and BLR- BTech-M Sc-MCA - II-IV and VI Sem - End Semester Examination - Apr-May 2025		ECE-2222:Digital Signal Processing	
#	Course Outcome	Description	Marks

No records found / CO Level Description not defined

* **NA** - CO Not Selected

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